

Team Eclipse

AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies

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Abstract—India imports roughly 88% of its crude oil, and nearly half of that volume passes through the volatile Strait of Hormuz. Recent geopolitical shocks — from the 2025 US-Iran stand-off to the ongoing Houthi attacks in the Red Sea — have repeatedly forced Indian refiners onto the costly spot market. With only 9.5 days of cover in India's Strategic Petroleum Reserve (SPR), a sustained supply disruption poses a serious economic risk. Traditional supply-chain tools are not built for this kind of volatility: they cannot dynamically model geopolitical effects, and they cannot automate rerouting, so recovery after a shock takes far longer than it should. We propose an AI-powered Energy Supply Chain Resilience System that shifts crisis response from a reactive scramble to a proactive, data-driven process. The system is built as a multi-agent pipeline purpose-built for energy security in a macroeconomic context:

- **Continuous Risk Ingestion** — a centralized ETL layer that aggregates continuous streaming data from across the globe: GDELT news signals, live vessel AIS tracking, OFAC/UN sanctions data, and market fundamentals.
- **Geopolitical Risk Intelligence** — an adversarial LLM debate between two opposing teams of threat-intelligence agents filters out noise to produce a calibrated disruption score for vulnerable chokepoints, along with human-readable risk summaries.
- **Dynamic Scenario and Impact Modeling** — a Knowledge Graph and Digital Twin represent the physical supply chain, while a Disruption Scenario Modeler uses causal graphs and Monte Carlo simulation to trace how a shipping delay would affect domestic fuel prices and GDP.
- **Automated Orchestration** — the pipeline produces actionable mitigation strategies: it mathematically optimizes the daily drawdown of the 9.5-day SPR to stabilize prices, and ranks alternative shipping corridors using procurement models.

All of these insights converge in a single Energy Resilience Command Center, giving decision-makers a real-time, actionable playbook to protect India's energy supply before a geopolitical event can damage the economy.

1. Introduction

Macroeconomic stability is closely connected to the resiliency of the energy value chain in India. Today, roughly 88% of the nation's crude oil comes from abroad and the flow is structurally insecure: nearly half of the imported crude oil transits through the volatile Strait of Hormuz. These trade routes are constantly at risk, ranging from the 2025 stand-off between the United States and Iran, to the ongoing Houthis' activities in the Red Sea. Indian refiners have to import fuel at high premiums on the volatile spot market when a route is blocked. India has only 9.5 days of cover in its Strategic Petroleum Reserve (SPR) and a sustained sea disruption would be not only a logistical headache, but an economic headache.

Legacy supply-chain management tools are simply not very useful in the face of geopolitical turmoil today. They're designed for predictable environments and are reactive: they can't dynamically model the cascading economic effects of a closed shipping lane, they can't automatically add unstructured geopolitical news to their picture of risk, and they can't automatically orchestrate procurement rerouting. This means the economy is vulnerable to prolonged fuel-price inflation and GDP drag following a geopolitical shock, and recovery takes a much longer time than it should.

We need to fill this intelligence gap by creating an 'Energy Supply Chain Resilience System' driven by an Artificial Intelligence system, thereby shifting the National crisis response from an 'after the crisis' scrambling to a mathematically informed and anticipatory process.

The proposed solution is a multi-agent data pipeline which is centered around four core operational layers, as depicted in the system architecture diagram in Fig. 1. The underlying ETL (Extract, Transform, Load) layer is Continuous Risk Ingestion that continuously streams unstructured data from around the world such as news events from the GDELT project, real-time maritime AIS tracking, international sanctions and market fundamentals. The Geopolitical Risk Intelligence Agent applies an adversarial "Hawk/Dove" large language model (LLM) debate to evaluate threats when they are ambiguous and to filter noise out, resulting in accurate, human-readable disruption scores for key global chokepoints. A Knowledge Graph and Digital Twin modules simulate and map the actual physical reality of the supply chain, with a Disruption Scenario Modeler which tracks the exact impact a local shipping disruption will have on domestic fuel price changes and other broader impacts on the economy using causal graphs and Monte Carlo simulation. Lastly, an optimisation layer produces a ranked list of rerouting options to the procurement teams, with a mathematical model of the optimal daily drawdown of the 9.5-day SPR necessary to stabilize domestic prices.

All these specialised AI modules come together in an integrated Energy Resilience Command Center, transforming chaotic global data into an actionable playbook that helps decision makers defend India's energy infrastructure before a geopolitical event hits the economy.

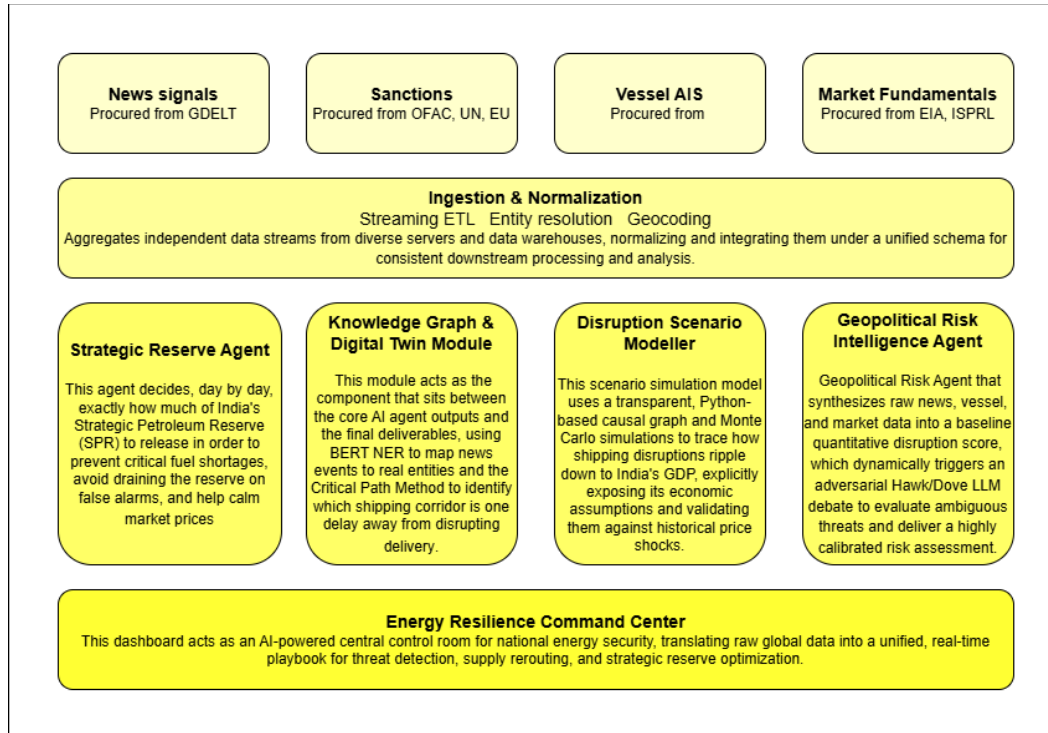


Fig. 1. Pipeline architecture of the proposed solution.

2. Ingestion and Data Scraping

2.1. Vessel AIS Ingestion

Vessel AIS Ingestion Agent is an agent that operates at the bottom of the proposed system architecture and offers the physical-tracking part of the system. It establishes a long-lived, reconnecting WebSocket connection to the real-time data stream at aisstream.io and provides a conversion of the unstructured maritime telemetry data to structured risk metrics. The world's position is narrowed to spatial bounding boxes of all the critical crude oil import routes into India (the Strait of Hormuz, Bab-el-Mandeb, the Persian Gulf and the Suez Canal), thus establishing the baseline routes to reroute through. The pipeline filters at the edge according to a strict class-based approach and allows all AIS ship-type codes 80-89 (liquid tankers and crude carriers) and secondary cargo codes 70-79. PositionReport and ShipStaticData messages are processed into a common spatial schema, and geofenced with a collection of pre-defined corridor polygons, for real-time tagging of vessel coordinates.

In addition to vessel tracking, the agent can also be used to serve as an automatic risk-intelligence engine to identify two types of abnormal activity related to vessels within a monitored chokepoint. First, it detects AIS-gap (“dark vessel”) events by highlighting any vessel that ceases transmission of AIS data for 30 minutes or more in a high-risk maritime corridor (transponder hushes) — which can be evidence of intended violation of sanctions, unmonitored ship-to-ship transfers, or simply deliberate attempts at evasion in the vicinity of kinetic conflict points. Second, it monitors the velocity profile of a vessel over time and alerts for chokepoint loitering when a tanker has been traveling at less than 1 knot for a prolonged period and is located in a narrow waterway, which could result from physical obstructions, or a tactical wait during an elevated threat scenario. All the normalized telemetry, spatial tags and anomaly flags are available in real-time in a central event data base that is queryable by the rest of the platform.

The agent is designed to overcome the cold-start challenge posed by live streaming feeds by incorporating an automated baseline-backfill pipeline that leverages IMF PortWatch’s daily transit datasets. Each monitored maritime corridor gets 365 days of daily vessel-throughput and crude-carrying-capacity information from this historical pipeline. These lagging historical baselines are compared to the live historical daily tanker counts from aisstream.io and the system calculates the rolling density z-score. If a statistical significance reduction in the daily number of vessels is identified, then it automatically triggers a quantitative disruption alert, as the empirical monitoring methods used to detect large-scale global supply-chain disruptions.

2.2. *Market Fundamentals Intelligence*

It also features a multi-source ingestion pipeline designed specifically for market fundamentals and government petroleum statistics, allowing for continuous threat detection and scenario modeling throughout the national energy supply chain. This component collects, normalizes and processes signals from raw economic and physical-market data sources and in real-time structure these signals into market stress scores, demand-pressure trends, import nowcasts, etc.

A. Heterogeneous Source Collection and Data Provenance

The ingestion layer is an ongoing link to official government and international energy-statistics data, which is received via special collectors. The real-time spot-price series for the international benchmark crudes: Brent (RBRTE) and West Texas Intermediate (RWTC) are sourced from the U.S. Energy Information Administration (EIA v2) API by International Spot Markets (EIA). National Trade & Consumption (PPAC) combines four key metrics from the monthly oil official report issued by the Petroleum Planning & Analysis Cell, the price of the Indian Basket crude and extracts them at a monthly frequency using a dedicated PDF parser. Strategic Reserves Reference (ISPRL) gets

data for inventory capacity from Indian Strategic Petroleum Reserves Limited, and then fills it with Press Information Bureau releases to maintain the reference values. No operational data is hardcoded in the pipeline, maintaining auditability and eliminating static assumptions. Each data element that is processed or generated by the system is clearly marked as either OFFICIAL (published government data), OBSERVED (direct fetch from the market price), NOWCAST (an estimate for a period where no government data is available), DERIVED (an indicator which is calculated from other data), or SCENARIO_STATE (a value assigned during the simulation process)

B. Parsing, Frequency Alignment, and Feature Engineering

There are inherent challenges in raw government trade data: daily spot-market prices have a daily frequency, while monthly petroleum reports have a monthly frequency. There is a strict multi-stage process in the preprocessing pipeline that overcomes that. PPAC parsing and validating: The PPAC parser supports units (MM is Million Metric Tonnes, MMT), and the parsing is anchored to a calendar report month to prevent year-to-date issues. Frequency alignment: The daily benchmark price series is normalized and cleaned, with two views created: historical aggregates (over full months) for model training and rolling aggregates (month-to-date) for estimation of live periods. Behind the scenes, the price features derived using purely mathematical transformations include rolling 30-day volatility, momentum, and z-scores, Indian Basket-to-Brent crude spread, demand-growth features, and calendar-lag features.

C. Nowcasting Engine

The machine-learning Nowcasting Engine provides volume and throughput predictions for the current month prior to the official release of volumes and throughputs, generating a statistically sound estimate to fill in the lag associated with volume and throughput reporting. The engine tests multiple models including seasonal autoregressive integrated moving average with exogenous regressors (SARIMAX), tabular Extreme Gradient Boosting (XGBoost) and historical persistence baselines. Expanding-window (EW) is used to validate the models, training on the historical data of the year up to the forecast month to avoid data leakage and ensure realistic inference. Each candidate is then evaluated separately on the held-out historical data using its Mean Absolute Percentage Error (MAPE), and the one with the lowest error is used, otherwise a naive persistence baseline is used. In production, a readiness module ensures that sufficient recent price and fundamental inputs are available for generating any nowcast, otherwise returning an explicit error return flag instead of silently failing.

D. Market Stress and Fundamental Intelligence Synthesis

Nowcasts and price indicators become higher level intelligence signals for downstream agents in the final section of the market fundamentals pipeline. The Market Stress Index (0-1) is a weighted logistic z-normalised aggregate of 4 components of global crude-market stress: short-term price shock (30%) and 30-day rolling volatility (30%) and price momentum (20%) and the Indian Basket-to-Brent spread (20%). If the data of a component is stale (difference between the current time and the last update time is greater than 7 days), then the data will be removed, and the rest of the weights will be re-normalized dynamically. The Directional Signal Engine assigns activity a class from STRONGLY_CONTRACTING to STRONGLY_EXPANDING based on the comparison of the current nowcasts against the official baseline (three-month average). Market Regime Synthesis then overlays fundamental-activity trends and global market stress on two separate axes to create a synthetic market regime that has a deterministic, readable label (e.g., EXPANDING_WITH_ELEVATED_STRESS).

2.3. *Sanctions Intelligence Engine*

The Sanctions Intelligence Module is a part of the pipeline that sits downstream of the first ingestion layer, and is deterministic, legal-risk monitoring. It brings together public sanctions information from the US Office of Foreign Assets Control (OFAC), the United Nations (UN), and the European Union (EU) to create a rich, continually refreshed risk signals. As opposed to the probabilistic reasoning employed by the upstream LLM agents, this module is grounded in explicit, auditable rules as membership in the sanctions program is a legal fact and not a probabilistic inference. It feeds into the Geopolitical Risk Intelligence Agent, the Scenario Modeler and the Procurement Orchestrator.

A. Multi-Jurisdictional Ingestion, Resilience, and Schema Normalization

During geopolitical issues, it is common for government web endpoints to experience latency or downtime; the ingestion architecture has been designed to offer graceful degradation of service as opposed to a failure. The module continuously accesses raw XML data from the OFAC SDN List, the UN Consolidated List and the Financial Sanctions (FSF) Files of the EU on a 6 hourly basis. The network fetchers incorporate automated retries in case of temporary endpoint instability, primary/fallback URLs, and sophisticated disk caching, so that the system continues to function on recently cached data in case of temporary endpoint unreachability. The parsing layer fills these structural differences between the three authorities by converting each of the authorities' unique native XML schema representations into a common, single representation of NormalizedEntity. The standardized schema includes the primary name(s) of each entity, aliases, any related sanctions programs, entity type, listing

date, remarks, and — most importantly for this use case — any vessel International Maritime Organization (IMO) numbers associated with the entity.

B. Deterministic Energy-Relevance Scoring and Corridor Rollup

Every ingested entity is evaluated for energy relevance using a strict, rule-based system that is based on auditable criteria including being part of oil-linked sanctions programs, energy or tanker keyword matches in the remarks, and specific flags for vessel type. From these rules it dictates a structured relevance tag (CRITICAL, HIGH, MEDIUM, LOW or NONE). This logic will not rely on a blackbox LLM processing, so all scores will be fully testable, and will come with a human-readable array of `relevance_reasons` for operational transparency. The module then builds up tagged records at the corridor level by geography to create a risk signal (0-100) that ranks the corridor, for instance, the Strait of Hormuz or the Russia-India Direct route, according to the level of risks in that corridor, according to the number and strength of active and energy-relevant sanctions that are contained there. This is a "combined score," where the downstream Geopolitical Risk Agent can plug in legal restrictions as part of its overall disruption-probability calculations.

C. Continuous Delivery and Interoperability

The processed intelligence is provided via a live, auto-refreshing FastAPI service, avoiding the need for weekly database exports, as required. The API has several endpoints for performing cross module interoperability:

- **Quantitative Risk Polling** (/risk-signal) — polled continuously by the Geopolitical Risk Intelligence Agent to retrieve corridor-level metrics, including active sanctions counts, critically designated entities, and the composite risk-contribution score.
- **Counterparty Exclusion** (/entities) — queried dynamically by the Procurement Orchestrator to filter out sanctioned tankers or counterparties before ranking alternative crude sources and rerouting schedules.
- **Asynchronous Override** (/refresh) — a manual trigger endpoint that forces an immediate, out-of-cycle data pull when upstream news agents detect a breaking sanctions announcement, e.g. via webhook.

By storing each vessel's `imo_number` attribute in the `NormalizedEntity` schema a direct join key with the Vessel AIS Ingestion Agent is created. This exact-match capability lets the Command Center link deterministic legal sanctions with spatial telemetry, enabling real-time tracking of shadow-fleet vessels transporting sanctioned crude.

2.4. GDELT Event Ingestion

To complement maritime AIS telemetry and quantitative market signals, the system incorporates a dual-stream geopolitical event-ingestion pipeline. This engine continuously ingests unstructured geopolitical and conflict-event data from two primary global intelligence feeds — the Global Database of Events, Language, and Tone (GDELT). It normalizes both heterogeneous sources into a unified common schema, tags events to spatial chokepoints, and loads records into Google BigQuery for real-time querying by downstream risk-scoring models.

A. Dual-Source Ingestion Modes and System Architecture

The ingestion pipeline combines server-side SQL queries with client-side REST polling to pull global conflict data. Server-side GDELT ingestion queries GDELT records directly within Google BigQuery, referencing the public dataset `gdelt-bq.gdeltv2.events_partitioned`; this executes SQL transformations inside Google's cloud infrastructure and writes normalized records directly into destination tables without any local file downloads. REST API ACLED ingestion is used because ACLED has no native BigQuery mirror: events are fetched via ACLED's REST API, normalized locally with pandas, and loaded into BigQuery through structured dataframe jobs. A direct high-frequency hotpath is also available for lower-latency, “true real-time” ingestion: an optional script bypasses BigQuery's public-mirror delay by pulling GDELT's raw 15-minute update CSV dumps directly from `data.gdeltproject.org`.

GDELT (BigQuery Public Dataset) and ACLED (REST API Stream) → Common Schema Normalization → Corridor Geofencing → BigQuery (events_normalized)

B. Canonical Event Schema and Spatial Corridor Tagging

Regardless of whether an event originates from news scraping or human-coded conflict logs, it is transformed into a standardized 11-column relational schema before storage:

Canonical Schema = { event_id, source, timestamp, event_type, lat, lon, corridor_id, entities, severity_score, confidence, raw_reference }

Every ingested event is spatially evaluated against six predefined geographical bounding boxes corresponding to major crude-import chokepoints:

- Hormuz (Lat 24.5°-27.5°, Lon 54.5°-57.5°)
- Bab-el-Mandeb (Lat 11.5°-14.0°, Lon 42.0°-44.5°)
- Persian Gulf (Lat 24.0°-30.5°, Lon 47.5°-56.5°)
- Red Sea (Lat 12.0°-28.0°, Lon 32.0°-44.0°)
- Suez Canal (Lat 29.5°-31.5°, Lon 32.0°-33.0°)
- India Domestic / Coastal (Lat 6.5°-35.5°, Lon 68.0°-97.5°)

Events falling outside all six bounding boxes are dropped at the edge to conserve downstream analytical capacity.

C. Source-Specific Severity and Confidence Normalization

Because GDELT and ACLED record event severity and reliability on different scales, the pipeline applies source-specific mathematical transformations to normalize both `severity_score` and `confidence` onto a standardized [0,1] continuous scale.

GDELT normalization. GDELT severity combines article sentiment tone (`AvgTone`, bounded between -20 and $+20$) with event-impact intensity (`GoldsteinScale`, bounded between -10 and $+10$):

```
severity_score_GDELT = 0.5 × [(20 - clip(AvgTone, -20, 20)) / 40] + 0.5 × [(10 - clip(GoldsteinScale, -10, 10)) / 20]
```

GDELT confidence scales linearly with the number of independent news outlets covering the event (`NumSources`), capped at 10 corroborating sources:

```
confidence_GDELT = clip(NumSources / 10, 0, 1)
```

ACLED normalization. ACLED severity is derived directly from verified fatality counts, capped at a baseline threshold of 20:

```
severity_score_ACLED = clip(fatalities, 0, 20) / 20
```

Because ACLED events are manually coded and verified by expert analysts, confidence is set to a constant high baseline:

```
confidence_ACLED = 0.80
```

D. Append-Only Storage and Deduplication Architecture

To support serverless operational constraints and avoid costly Data Manipulation Language (DML) operations such as `MERGE` or `UPDATE`, the BigQuery storage layer uses an append-only architecture paired with a deduplicating view.

1. **Append-Only Base Table** (`events_normalized`): all ingestion threads load new and historical events using `WRITE_APPEND` dispositions. The base table is partitioned by date (`PARTITION BY DATE(timestamp)`) and clustered by `corridor_id` and `source` to optimize query latency and cost.
2. **Deduplicating Analytical View** (`events_normalized_view`): re-running historical backfills or overlapping live polling windows can introduce duplicate rows into the append-only base table. To guarantee idempotency, the system materializes a deduplicating view using an analytical window function.

```
SELECT * FROM `events_normalized` QUALIFY ROW_NUMBER() OVER (PARTITION BY event_id ORDER BY timestamp DESC) = 1;
```

3. Normalization and Unified Schema

Operating between the four heterogeneous Tier 1 data sources and the downstream AI agents (Geopolitical Risk Agent, Disruption Scenario Modeller, Procurement Orchestrator, and Strategic Reserve Agent), the Streaming ETL & Signal Unification Layer serves as the primary normalization and data routing backbone of the architecture. Rather than re-ingesting raw feeds, this layer continuously polls the structured outputs of the individual Tier 1 engines, transforming disparate data models into a unified schema stored across dual live-serving and model-training databases.

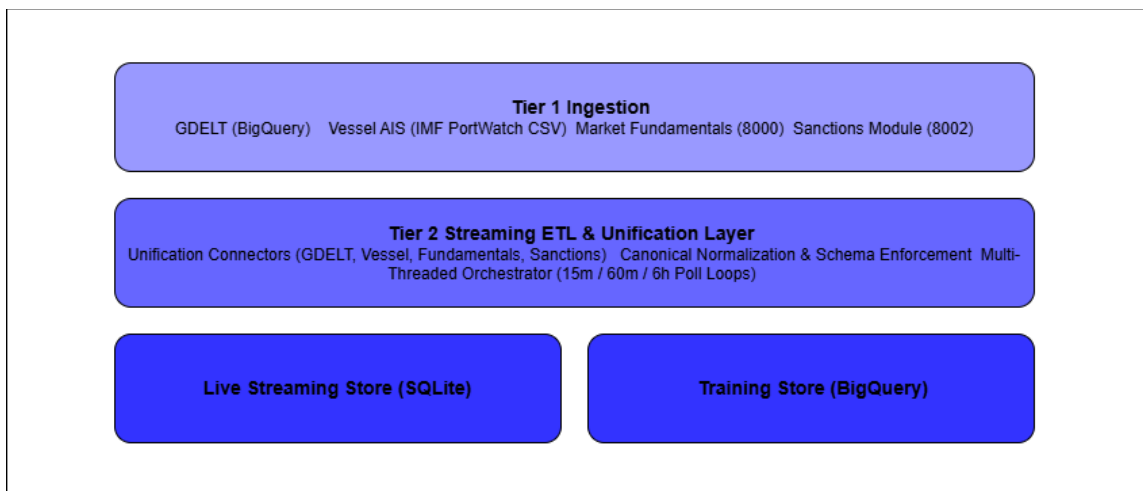


Fig. 2. Normalization stack.

The unification layer interfaces with four specialized Tier 1 source connectors:

1. **GDELT News Connector** — incremental checkpoint reader that queries newly ingested BigQuery rows written by the high-frequency event pipeline, transforming raw news severity into standardized signal records.
2. **Vessel Congestion Connector** — acts as a transitional ingestion engine by reading and extrapolating historical daily transit counts (e.g., IMF PortWatch data across Suez, Bab-el-Mandeb, and Hormuz) into live congestion metrics.
3. **Market Fundamentals Connector** — interrogates the local fundamentals REST API (Port 8000) and flattens nested economic JSON structures containing spot prices, PPAC consumption figures, and nowcast estimates into discrete rows.
4. **Sanctions Intelligence Connector** — polls the live Sanctions Module API (Port 8002) across three distinct endpoints (/risk-signal, /supplier-signal, and /alerts/new-listings), capturing legal risk metrics and active designation warnings.

To ensure that downstream agents consume a predictable, strongly typed interface regardless of signal origin, every connector normalizes incoming records into the canonical `CorridorSignal` object:

```
CorridorSignal(  
    corridor_id="hormuz",           # Geographic chokepoint or trade lane ID  
    as_of=datetime(2026, 7, 22, ...), # UTC timestamp  
    signal_type=SignalType.VESSEL_CONGESTION, # NEWS_EVENT | VESSEL_CONGESTION | MARKET_FUNDAMENTAL | SANCTIONS_RISK  
    metric="congestion_index",      # Derived metric identifier  
    value=0.73,                    # Normalized numerical measurement  
    unit="score_0_1",              # Unit of measure  
    data_status=DataStatus.SIMULATED, # Load-bearing provenance status  
    source=SourceSystem.PORTWATCH_SIM, # Originating subsystem  
    confidence=0.80,                # Confidence score [0, 1]  
    raw_reference="https://..."  # Traceability URI or source citation  
)
```

Fig. 3. Canonical CorridorSignal schema.

MODEL BUILDING

1. Scenario Simulator

The Scenario Simulator is an AI-assisted decision-support module that lets planners evaluate the impact of hypothetical disruptions across the global energy supply chain before they occur. Rather than only displaying current conditions, it performs what-if analysis by combining the live intelligence signals already produced upstream — vessel congestion, sanctions, geopolitical news, and market fundamentals — to estimate how a given disruption would affect procurement, transportation, and strategic reserves. In effect, it converts raw operational data into concrete scenarios that decision-makers can evaluate before committing to a course of action.

1.1 Objective

The simulator is designed to answer the kinds of questions a planner would otherwise have to reason through manually: What happens if the Suez Canal is blocked? What if sanctions are imposed on a supplier? What if geopolitical tensions escalate in the Strait of Hormuz? How would vessel congestion affect delivery schedules, and should procurement switch suppliers or draw down strategic reserves? Rather than waiting for a disruption to occur, the simulator lets planners anticipate its consequences and prepare mitigation strategies in advance.

1.2. Data Inputs

The simulator consumes normalized data produced by the Tier-2 integration layer described in Section 3, drawn from four sources.

A. Vessel Congestion

Derived from AIS-based shipping information, expressed as a congestion index alongside vessel count and tanker capacity for each corridor — for example, a congestion index of 0.65 and 46 vessels currently transiting the Suez corridor.

B. Sanctions Intelligence

Drawn from the OFAC, UN, and EU sanctions datasets described in Section 2.3, expressed as a corridor-level sanction risk score (for example, a score of 100 for Hormuz, indicating significant regulatory or geopolitical exposure for suppliers or routes through that corridor).

C. News Intelligence

Drawn from GDELT, expressed as an event-severity score (for example, 0.81 for Hormuz), where a higher score indicates greater geopolitical instability.

D. Market Fundamentals

Where available: crude price, demand index, and the PPAC and EIA statistics described in Section 2.2, which supply the economic context against which a scenario is evaluated.

1.3. Simulation Workflow

The simulator follows a structured, four-step pipeline: it retrieves live corridor signals, applies the selected disruption scenario, estimates the resulting impact, and generates recommendations.

Step 1. Collect Current Intelligence

The simulator retrieves the latest corridor signals from the Tier-2 backend (via a /corridors/latest endpoint) for each monitored corridor — Suez, Hormuz, Bab-el-Mandeb, the Russia-India route, and the Venezuela-India route — each carrying its current congestion, sanctions, news severity, confidence, tanker capacity, and vessel count.

Step 2. Scenario Selection

A planner selects a disruption scenario — for example, a Suez Canal closure, a Hormuz conflict, a Russian oil embargo, a high-congestion event, or Red Sea attacks — and the simulator modifies the relevant corridor conditions accordingly.

Step 3. Impact Estimation

The simulator then evaluates how the selected scenario propagates across four dimensions. Transportation impact: higher congestion translates into longer transit times, larger vessel queues, and reduced throughput. Procurement impact: elevated sanctions risk flags suppliers that require review or replacement. Geopolitical impact:

high GDELT event severity signals increased instability, a higher disruption probability, and greater operational uncertainty. Reserve impact: when multiple disruptions occur simultaneously, the simulator estimates whether strategic reserves would need to be drawn down.

1.4. Decision Logic and Outputs

The simulator combines these indicators through a small set of interpretable rules — for instance, congestion above 0.60 together with sanctions above 80 is classified as high supply-chain risk; news severity above 0.75 triggers a recommendation to increase monitoring; and congestion above 0.60 on its own triggers a recommendation to use an alternate route. These rules allow rapid, auditable generation of operational recommendations, which the simulator packages into five structured outputs for planners: an overall risk assessment (e.g., HIGH); procurement advice (e.g., avoid procurement through Hormuz and consider alternative suppliers); a route recommendation (e.g., avoid Suez, use the Cape of Good Hope); a reserve recommendation (e.g., maintain an additional five-day inventory buffer); and a scenario summary combining the scenario name, expected delay, and overall risk rating.

1.5. Integration and Current Status

The Scenario Simulator sits directly downstream of the Tier-2 integration layer described in Section 3, consuming the same live data already surfaced on the operational dashboard and using it to evaluate hypothetical situations rather than only the current state. It provides early identification of supply-chain disruptions, evaluation of multiple hypothetical scenarios in parallel, and decision support for procurement planning, inventory management, route selection, and geopolitical awareness. At the current stage of implementation, Tier-2 data integration, live congestion data, live sanctions data, and live GDELT news severity are all integrated, and the dashboard displays current intelligence with the scenario logic already prepared to consume these live signals; more advanced optimization — dynamic rerouting, a full digital twin, or machine-learning-based simulation — remains a planned future iteration rather than part of the current build.

2. Geopolitical Risk Intelligence Agent

The Geopolitical Risk Intelligence Agent is a dual-tier disruption-risk scoring system for India's crude-oil import corridors — Hormuz, Bab-el-Mandeb, Suez, the Persian Gulf, and the Red Sea. It combines an always-on quantitative layer with an adversarial LLM debate that is triggered only for ambiguous or elevated cases, ingesting GDELT

news, AIS-style vessel traffic, commodity fundamentals, and sanctions data to produce a single, per-corridor, per-timestamp disruption-probability score.

2.1 Pipeline Architecture

Raw GDELT news is first deduplicated, keyword-filtered, and semantically rescued via TF-IDF similarity before being reassigned to the correct corridor. This filtered corpus, together with vessel-history and fundamentals data, feeds a Tier-1 quantitative stage that produces separate news, vessel, and fundamentals sub-scores, combines them with a sanctions multiplier, and outputs a composite Tier-1 score for every corridor at every timestamp — explicitly marking any corridor lacking sufficient history as NO_DATA rather than substituting a fabricated zero. A Tier-2 retrieval stage then builds a credibility-tiered evidence store over the filtered news corpus using TF-IDF vectorization, so that any escalation to the LLM layer is grounded in retrievable, quoted evidence rather than free-form generation. A Tier-3 stage detects trigger conditions, collapses related events into episodes, and runs a two-round adversarial Hawk/Dove debate — independent scoring, then rebuttal — followed by an Adjudicator that reconciles the two positions and applies guardrail logic such as a credibility floor on the more alarmist (Hawk) position. A Tier-4 stage mines historical backtest candidates, computes a lead-time metric against Brent price movements, and sweeps trigger thresholds to calibrate the system; Tier-5 applies this calibration and merges the Tier-1 and Tier-2 outputs onto a single schema, producing a unified risk feed and a ranked disruption-alerts output that downstream agents — including the Scenario Simulator, the Knowledge Graph module, and the Strategic Reserve agent — consume directly.

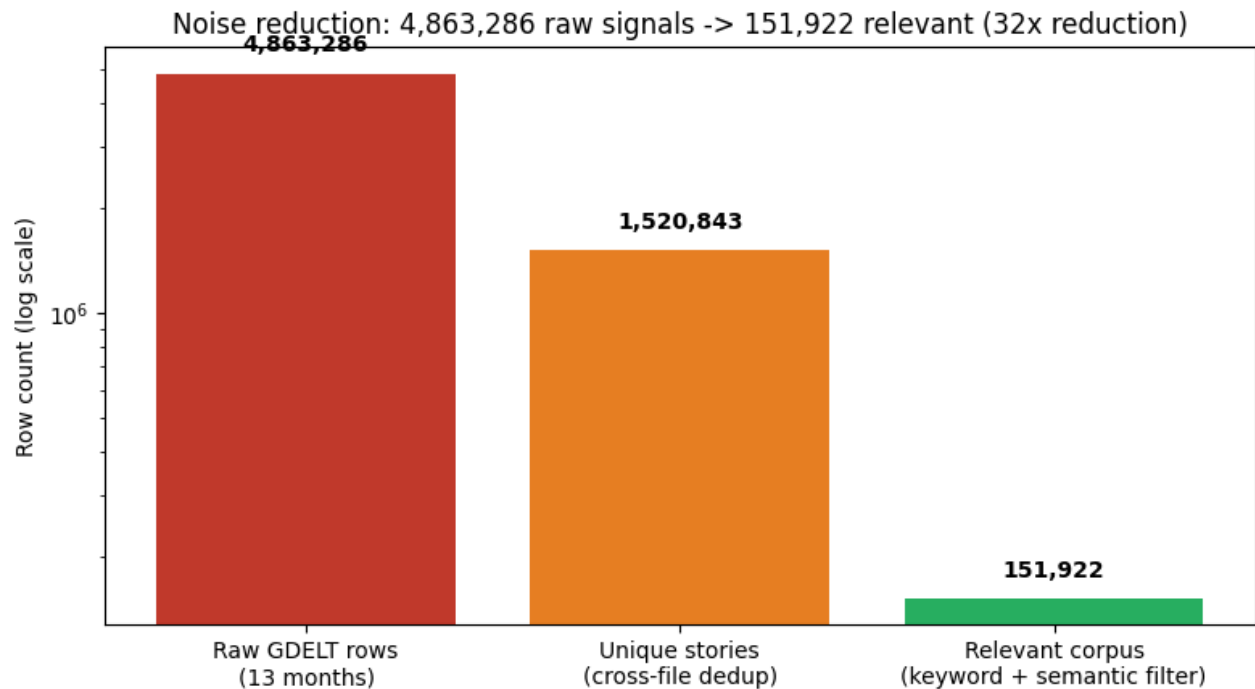


Fig 4. Noise reduction pipeline for the GDELT news dataset, illustrating a 32x compression from 4.8 million raw signals to a filtered corpus of 151,922 relevant stories.

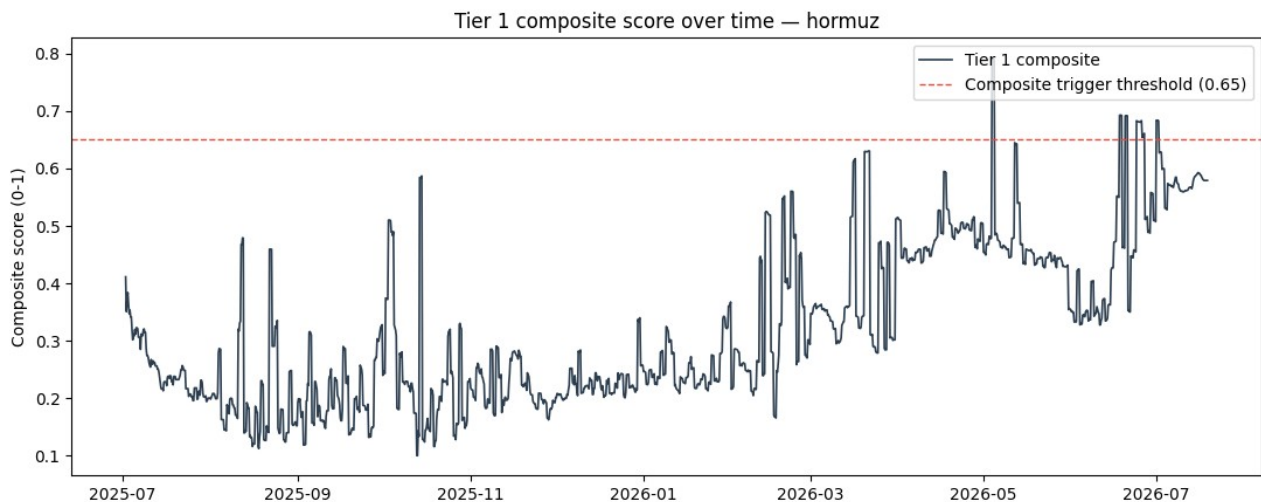


Fig 5. tier-1 composite risk score for the Strait of Hormuz over a 13-month backtest. Spikes that cross the 0.65 composite trigger threshold (dashed red line) indicate periods where the adversarial LLM debate is activated.

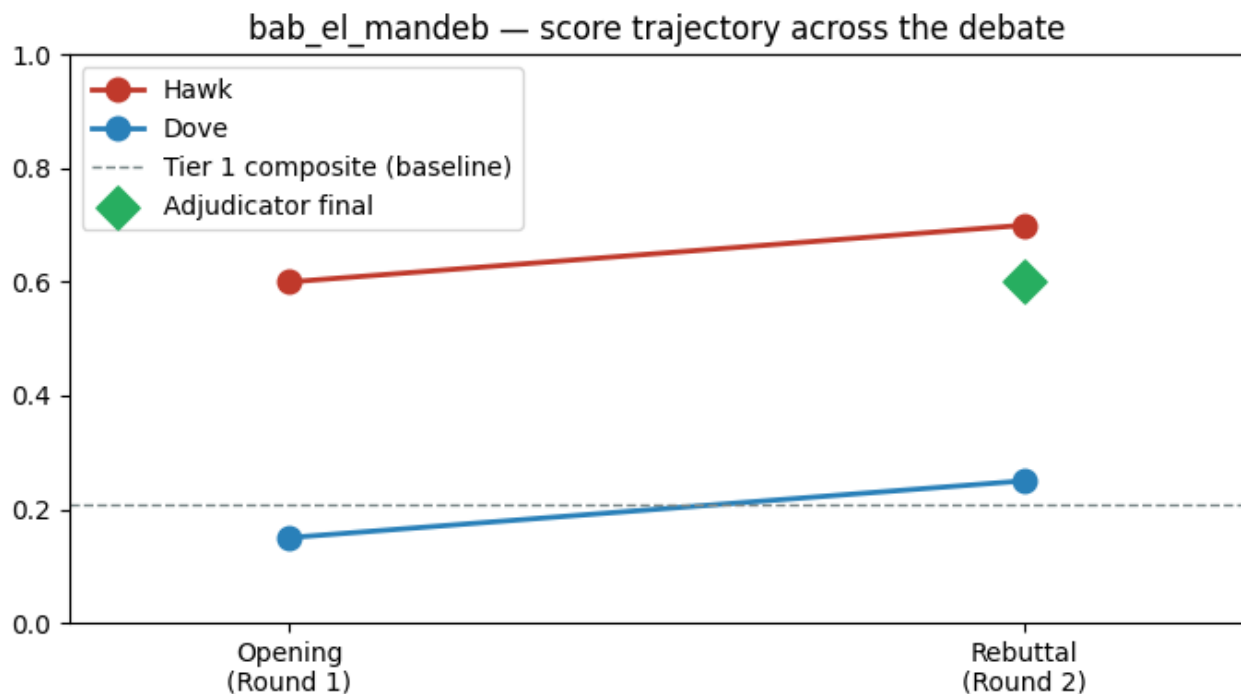


Fig 6. Internal mechanics of the LLM escalation layer. Despite the Dove arguing for a score near the Tier-1 baseline, the Adjudicator's guardrail logic ultimately sides closer to the alarmist (Hawk) position following the Round 2 rebuttal.

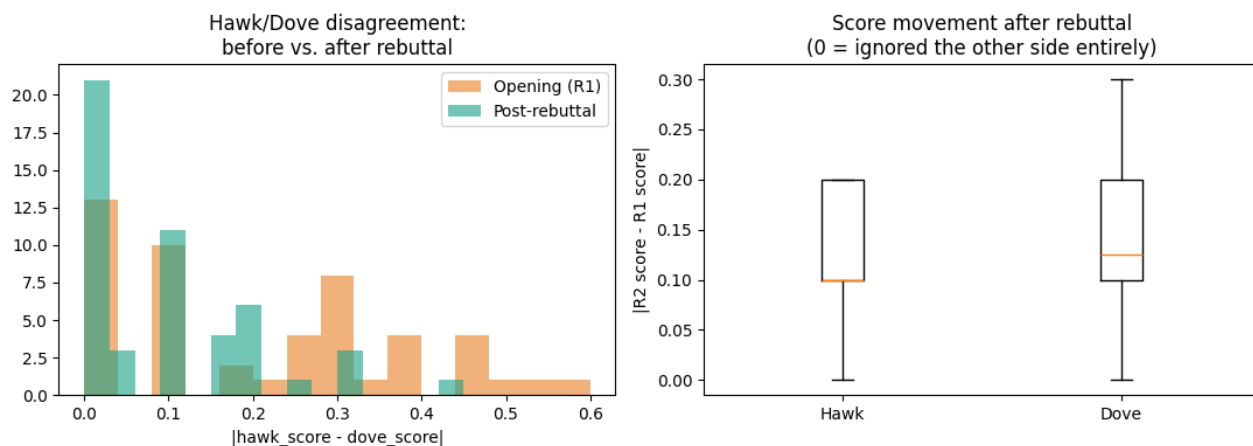


Fig 7. Hawk/Dove score convergence metrics. The data illustrates that the mandatory rebuttal round successfully reduces inter-agent disagreement, with both models demonstrating measurable flexibility in their final scoring.

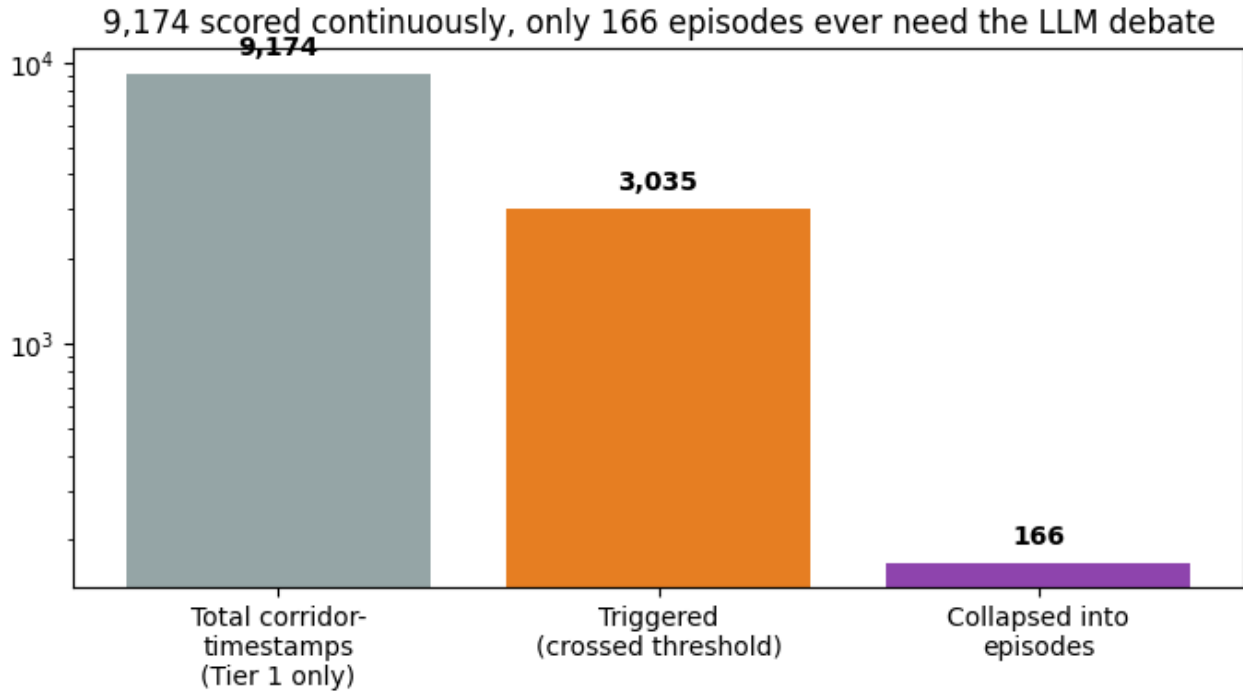


Fig 8. Volumetric decay of risk events prior to LLM escalation. The chart highlights the critical role of the episode-collapse mechanism, which compresses 3,035 threshold-breaching timestamps into just 166 unique geopolitical episodes requiring Adjudicator review.

2.2. Output Schema

Each row of the unified risk feed corresponds to one corridor at one scoring timestamp, and reports which tier actually produced the number:

Field	Meaning
corridor_id	Hormuz, Bab-el-Mandeb, Suez, Red Sea, Persian Gulf, or India
tier1_composite_score	Always-on quantitative composite (NaN / NO_DATA when insufficient history exists)
tier2_triggered / tier2_scored	Whether the trigger condition fired, and whether Tier 2 was actually run for this row
hawk_score / dove_score	Round-2, post-rebuttal scores from the adversarial debate
adjudicator_final_score, disagreement_index, human_review_flag	Adjudicator output and escalation flag
effective_score /	The single number downstream agents should use, and whether

Field	Meaning
score_source	it came from Tier 1 alone or from Tier 2 adjudication

2.3. Calibration and Validation

Calibration against the backtest data showed the composite trigger threshold (0.65) had little effect across a 0.55–0.75 sweep, but that the disagreement threshold needed to rise from 0.35 to 0.45, with fundamentals excluded from the disagreement calculation — a flat monthly fundamentals series was mechanically disagreeing with daily news and vessel sub-scores and inflating the trigger rate for structural rather than risk-related reasons. This reduced the observed trigger rate from 33.1% to 1.6% of all corridor-timestamps, meaning the expensive LLM path now runs on a small fraction of cases. The human-review threshold was correspondingly lowered from 0.30 to 0.20, since the actual disagreement_index distribution never exceeded roughly 0.205 at the 95th percentile and the original 0.30 threshold flagged none of the twenty real backtest episodes. The price-trigger logic was independently validated against a real event: it correctly fires in March 2026, when Brent rose to \$126.69 from \$71.32 in February — the one genuine Brent shock present in the dataset.

2.4. Scope and Limitations

The agent deliberately does not include a Knowledge Graph, Procurement Orchestrator, or Reserve Optimization logic of its own — those require supplier, refinery, route, and SPR data that this module does not ingest, and are instead handled by the dedicated modules in Sections 6 and 7, which the unified risk feed is designed to plug directly into. Backtest candidates are currently mined but not hand-labeled, so the lead-time metric remains a scaffold rather than a validated ground truth. The injection-resistance check is only half-closed: credibility tiering has been shown to suppress a fabricated low-credibility story in the retrieval layer, but the corresponding live Dove-suppression test has not yet been run. The retrieval and rescue layers use TF-IDF similarity as an explicit, labeled proxy for a real embedding model, and the sanctions multiplier remains a static, live-only rule rather than a historical time series, since no historical OFAC/UN backfill exists.

3. Knowledge Graph and Digital Twin Module

The Knowledge Graph and Digital Twin module sits between the four upstream agent outputs — Sanctions, GDELT, Vessel/AIS, and Market Fundamentals — and the final deliverables of the system: disruption alerts, reroute plans, and SPR briefs. It represents the supply chain as a supplier-route-refinery graph and layers a geospatial

simulation on top of it, so that abstract risk scores can be translated into a concrete statement about where a delay would actually propagate.

3.1. Core Techniques

Two complementary techniques do the work. A pretrained BERT named-entity-recognition (NER) model reads GDELT news text and confirms whether a mention such as “Hormuz” genuinely refers to a real place or entity before a gazetteer maps it onto the corresponding graph node — replacing plain keyword search with typed entity detection, and requiring no training data since the model is used as-is. The Critical Path Method (CPM), borrowed from project scheduling, then runs over the resulting graph: every corridor's transit time is inflated by its live risk score, and CPM's forward and backward pass identifies which corridor has zero slack — that is, which corridor's delay would directly delay India's crude arrival. This approach is fully deterministic and explainable, which matters given that the project has no labeled historical disruption events available to train a graph model against.

3.2. Illustrative Result

In a test run, the highest-risk corridor — the Strait of Hormuz, at a risk score of 0.54 driven by real vessel congestion, sanctions, and news signals — was not the corridor CPM identified as critical. The route via the Cape of Good Hope was critical instead, because it is the longest physical voyage and has zero absorption margin regardless of its risk score. This distinguishes two genuinely different, complementary questions: which corridor is riskiest, which the Geopolitical Risk Agent already answers, versus which corridor is one delay away from disrupting delivery, which is what CPM adds.

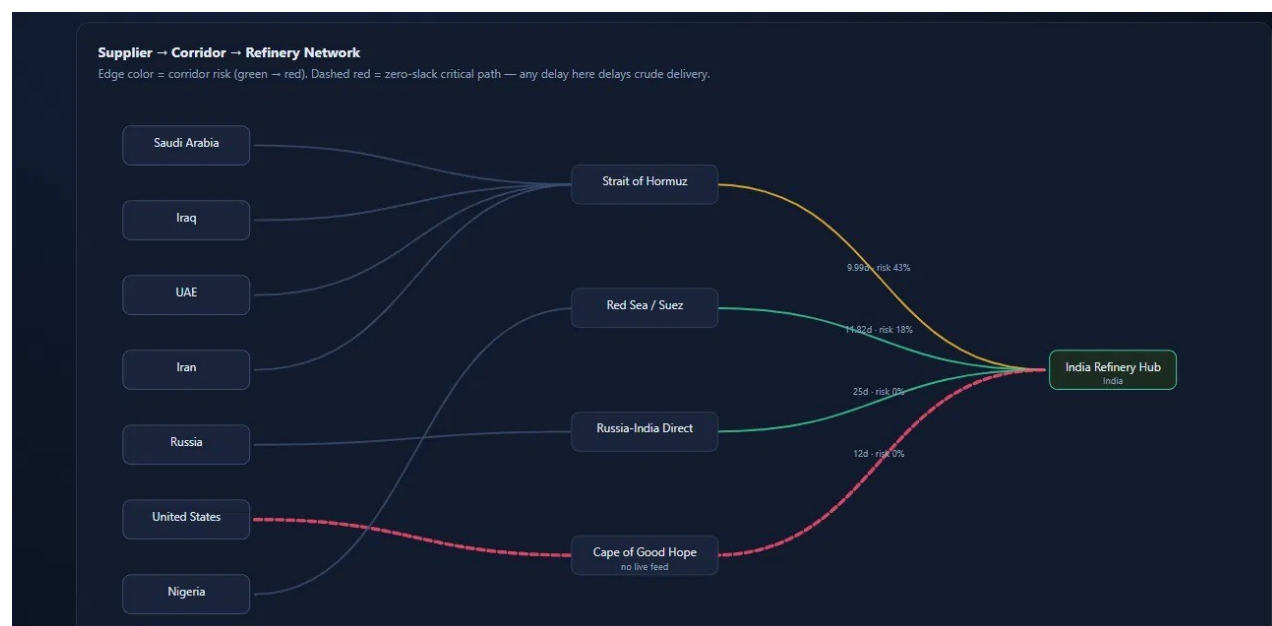


Fig 9. A supplier-route-refinery graph applying the Critical Path Method (CPM) to evaluate delay propagation.

3.3. Interfaces and Data Requirements

The module exposes its outputs through a small FastAPI service (/graph, /graph/geojson, /critical-path, and /corridor-slack), and consumes vessel data keyed by chokepoint and either transiting deadweight tonnage or vessel count, GDELT text for entity extraction, and sanctions data keyed by energy relevance and corridor or program — including, where available, the corridor-level output of the sanctions module described in Section 2.3. Market-fundamentals data is loaded but intentionally not yet consumed here, since Brent/WTI and consumption series feed the Scenario Simulator and Strategic Reserve agent more directly than this module's transit-time model.

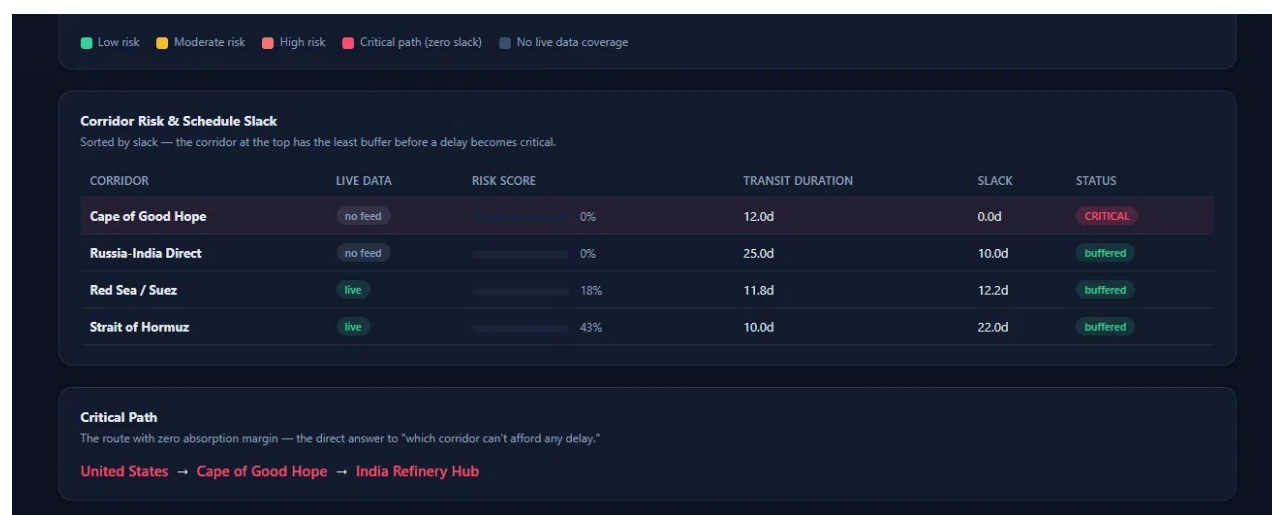


Fig 10. A tabular view distinguishing overall risk from delay vulnerability. The Cape of Good Hope is identified as the zero-slack critical path despite having a lower risk score than the Strait of Hormuz.

3.4. Limitations

Corridor transit times are public shipping-distance estimates rather than a live AIS route-time series, and are flagged as an explicit assumption rather than hidden. The BERT overlay currently produces only a corridor-level average risk score from GDELT mentions, without per-event recency decay — a natural next step given more time. Fundamentals data remains loaded but unwired into the graph logic by design, pending integration with the Scenario Simulator and Reserve agent.

4. Strategic Reserve Optimization Agent

India's Strategic Petroleum Reserve (SPR) is an emergency stockpile of crude oil sized to cover roughly 9.5 days of national fuel use if imports are disrupted — for example, by a closure of the Strait of Hormuz. This agent decides, day by day, how much of that reserve to release, so that the country never runs critically short during a genuine supply shock, the reserve is not drained unnecessarily on false alarms, and the release schedule can help stabilize prices rather than only cover a physical shortfall. Its output is a day-by-day drawdown schedule — release volume, remaining reserve, and days of cover — together with the reasoning behind each day's decision, feeding into the same dashboard as the Risk, Scenario Simulator, and Procurement agents.

4.1. *Approach*

Most public SPR models, including those used by the IEA and EIA, apply one deterministic rule: release today whatever the physical shortfall is today. This agent instead combines two techniques. Model Predictive Control (MPC), a standard technique from robotics and power-grid balancing not previously applied to reserve scheduling in any public model the team could find, re-solves a small optimization problem every day over a rolling 14-day look-ahead window and commits only to that day's action, so it reacts to new signals as they arrive rather than following a fixed plan. A dual-objective trigger then decides when to act, watching two independent signals: a physical trigger, based on whether vessel traffic is abnormally low right now at any of the three real Indian crude-import chokepoints (Hormuz, Bab-el-Mandeb, Suez) relative to that corridor's own recent history — the worst-affected corridor on a given day drives the decision, since a Suez disruption matters as much as a Hormuz one — and a price trigger, based on whether Brent crude is spiking relative to its own recent history. Physical need is always a hard requirement that cannot be skipped; the price response is an additional bonus release during a price spike, automatically disabled once the reserve nears its safety floor so it can never be the reason the country runs genuinely short.

4.2. *Backtest Results*

Backtesting against 357 days of real data across all three corridors compared the MPC-based agent to a naive baseline that reacts only to the current day's shortfall:

Metric	MPC agent	Naive baseline
Total volume released	5.36 MMT	3.25 MMT
Minimum cover reached	1.10 days	4.42 days
Days with any release	95	81

Of the 81 physical triggers, Hormuz drove 40, Bab-el-Mandeb 22, and Suez 19 — confirming that Bab-el-Mandeb and Suez each contribute genuinely independent trigger events, and that over a third of real triggers would have been missed by a Hormuz-only model. The MPC agent releases more and allows cover to run lower because it is actively using reserves for price stabilization as well as physical shortage — 29 of its release days were price-bonus-driven, something the naive model cannot do at all — which is a deliberate design trade-off rather than a shortcoming. As a calibration check against real ground truth, the price trigger correctly fires in March 2026 (\$126.69, up from \$71.32 in February) — the one genuine Brent shock present in the dataset.

4.3. *Replenishment Modeling*

A further extension models the refill side of the reserve lifecycle as a pipeline delay rather than an instant top-up: a VLCC lead time of 37 days (the midpoint of a cited 30–45 day range), an order size capped at one VLCC cargo (about 0.273 MMT), orders placed only during calm periods with a three-day cooldown between them, and a storage cap at the reserve's original starting level. Modeling this refill pipeline explicitly — rather than assuming reserves silently reset — cut the number of days spent below the three-day safety floor from 85 to 8 across the 357-day backtest, and raised the minimum cover level reached from 1.10 to 3.52 days. Forty-five refill orders were placed (6.90 MMT total), of which thirty-nine (5.94 MMT) had landed by year-end. This is the clearest evidence in the backtest that a resilience model ignoring replenishment lead time understates real exposure to a second, closely following disruption.

4.4. *Assumptions and Limitations*

The model fixes a safety floor of three days of cover that is never touched by discretionary releases, a 14-day MPC look-ahead horizon, a disruption-decay constant of 1.4 days measured directly from the data, and a maximum daily release capped at 3% of the remaining reserve. The VLCC lead time and cargo size are real-world reference figures rather than values confirmed against the specific fleet used, and are flagged as assumptions on the same basis as the other configuration parameters. Replenishment ordering currently follows a simple cooldown-and-target-gap heuristic rather than its own optimizer; a natural next step would fold refill timing into the same MPC objective rather than triggering it as a separate rule. Notably, the price-bonus logic initially had no effect at all, then over-corrected and drained the entire reserve, before reaching its current, validated behavior — evidence that the model was iterated against real data rather than only theorized.